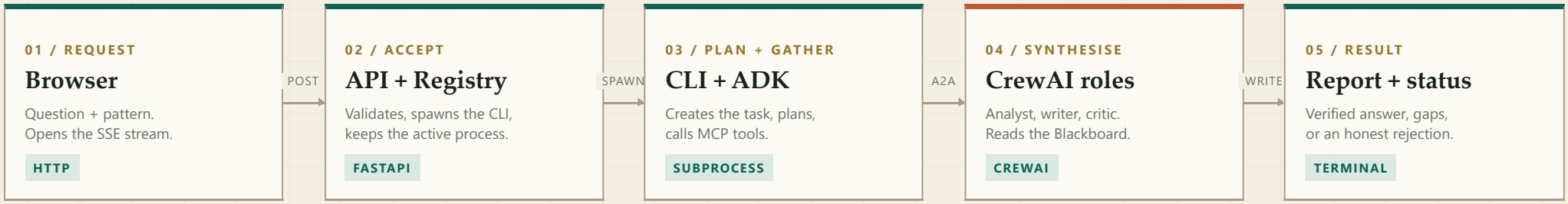
ONE QUESTION, FOUR BOUNDARIES

From question to verified answer.

A browser starts the run. ADK plans and researches. CrewAI writes and reviews. A shared Blackboard carries the evidence, while a separate event log carries the trace.

Control path

The browser never calls a model, a tool, or the provider directly. The local API starts the run and reads the stores.



BLACKBOARD / RESEARCH STATE

Task → plan → evidence → analysis → draft → final report → critique.
`data/blackboard.db`
Written by ADK and CrewAI; read by every stage and by the API snapshot.

EVENT LOG / TRACE

Stage, model, tool, A2A, limiter, error and `run_summary` events.
`data/events.db`
Feeds the timeline, metrics, heatmap and SSE stream.

RUN FILES

Process output and the final summary JSON.
`.run/runs/{id}.log`
`.run/runs/{id}.summary.json`
Read by the `/log` watchdog.

READ PATH `SSE /api/runs/{id}/stream` events + Blackboard snapshot → `/log` process exit / watchdog → Browser timeline, report, metrics

Evidence is the product.

Search finds candidates. Fetching produces source text. Only recorded, relevant evidence reaches the analyst, writer and critic.

Question
01

Explicit requirements: finding count, official sources, practical impact.

Route
02

External → web + fetch. Local → RAG. Mixed → both. Calculation only when the task needs it.

MCP tools
03

web_search returns discovery hints. fetch_page returns primary text. rag_search returns local chunks. calculate returns exact values.

RECORD

THEN

READ

→

EvidenceItem
04

Every item keeps id, source type, source ref, content and retrieved_by. Full content stays on the Blackboard.

Prompt evidence
05

Fetched pages get 2,000 chars; snippets, RAG and calculations get 600. Failed/empty fetches are excluded from citable evidence.

Analysis → Draft → Critique
06

The writer cites source refs. The critic checks claims against evidence. Rejection is a result, not a failure to hide.

SNIPPET ≠ EVIDENCE

A search result is a pointer. A claim that needs verification must rest on fetched page text.

FAILED FETCH ≠ EVIDENCE

Errors and empty pages remain visible for audit, but never become citable support.

MISSING EVIDENCE ≠ INVENTION

The writer states the gap. The reader gets an honest incomplete answer, not a fabricated one.

UNTRUSTED BY DEFAULT Fetched web text is wrapped as data. It cannot change the role, policy or tool rules. · **PROVENANCE** Every citable item carries a source_ref and the role that retrieved it.

Four patterns. One contract.

All four produce the same Blackboard sections and comparable reports. The difference is how work and context travel.

Sequential

01

One researcher with all tools. The baseline.

planner → researcher all tools → analyst → writer → critic

Parallel

02

Role-specific researchers run concurrently where planned.

planner → rag → web → calc → analyst → writer → critic

Hierarchical

03

Plan roles are dispatched in order; one worker run per unique role.

planner → unique workers in plan order → analyst → writer → critic

Recovery: high rejection + budget → targeted research → writer → critic again (bounded).

P2P

04

Context travels inline in A2A messages; the supervisor audits afterward.

planner message → researcher message → analyst inline → writer inline → critic inline

Difference: no agent reads the Blackboard during reasoning; the supervisor writes audit records.

Recovery, then honesty.

A critic rejection is useful only if it produces either better evidence or a clear final gap. It never becomes an approval.

Writer
Draft + structured FinalReport

→

Critic
Claim / source / completeness check

→

Approved
Every claim traceable; answer complete

Rejected, high, hierarchical
Targeted research gap → new evidence → writer + critic rerun, bounded

→

Rejected / no safe loop
Terminal rejected or incomplete result; gaps stay explicit

Approved
Critic approved and evidence sufficient.

Degraded
Stage error, warning, or partial evidence.

Rejected
Critic found concrete issues.

Failed
Execution stopped before a result.

Unconfirmed
Process ended without completion evidence.

Missing critique is not approval.
A missing or failed critic produces an incomplete result, never a verified one.

Disconnect is not success.
The SSE stream and /log watchdog only mark success on valid completion evidence.

Evidence gaps stay visible.
The report may be shorter than requested; it may not be invented.